

MINNEAPOLIS CRYSTAL, MN, USA

WMO: 726575

Lat: 45.062N

Lon: 93.351W

Elev: 262

StdP: 98.21

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 94960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.1	(c) -21.2	(d) -27.9	(e) 0.3	(f) -22.5	(g) -26.2	(h) 0.4	(i) -20.7	(j) 10.5	(k) -9.2	(l) 9.3	(m) -9.2	(n) 3.1	(o) 300	(p) 0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	32.5	22.9	31.1	22.2	29.1	21.0	24.8	30.3	23.6	28.8	22.5	27.6	5.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.9	18.2	28.3	22.0	17.2	27.4	20.9	16.1	26.1	76.6	30.2	71.9	28.8	67.5	27.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.5	(c) 7.7	(e) -27.2	(f) 35.5	(g) 2.8	(h) 1.7	(i) -29.2	(j) 36.7	(k) -30.8	(l) 37.7	(m) -32.4	(n) 38.6	(o) -34.4	(p) 39.9		
(a) 9.4	(b) 8.5	(c) 7.7	(e) -27.4	(f) 26.6	(g) 2.7	(h) 1.4	(i) -29.4	(j) 27.6	(k) -30.9	(l) 28.5	(m) -32.5	(n) 29.3	(o) -34.4	(p) 30.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.9	-8.8	-6.8	0.2	8.0	14.7	20.1	23.2	21.5	17.3	9.3	1.6	-5.9
	DBStd	12.36	6.91	6.69	6.81	5.63	4.79	3.80	3.12	3.07	4.73	5.18	5.73	6.50
	HDD10.0	2306	583	470	312	101	12	0	0	0	3	77	257	492
	HDD18.3	4197	841	703	563	311	134	25	2	7	75	284	502	751
	CDD10.0	1556	0	0	8	42	157	302	408	357	222	55	6	0
	CDD18.3	406	0	0	1	2	21	77	152	105	44	4	0	0
	CDH23.3	3709	0	0	3	29	236	687	1464	879	373	37	0	0
	CDH26.7	1179	0	0	0	6	76	208	522	256	105	7	0	0
Wind	WSAvg	3.6	3.7	3.7	3.8	4.1	3.8	3.4	3.2	3.0	3.4	3.6	3.7	3.6
Precipitation	PrecAvg	813	23	20	44	76	104	122	116	108	85	71	45	28
	PrecMax	1066	50	41	93	199	231	225	254	190	216	162	138	62
	PrecMin	593	2	4	10	24	5	45	28	10	10	15	2	7
	PrecStd	109	12	10	20	39	52	46	53	48	53	41	36	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.3	12.5	21.6	27.2	32.5	33.6	35.5	33.3	32.2	27.5	19.5	10.2
		MCWB	3.5	8.0	14.1	16.7	20.4	23.0	24.7	23.2	22.1	18.7	12.9	7.9
	2%	DB	3.9	7.4	17.0	22.6	28.7	31.1	32.6	31.3	29.4	23.6	16.2	7.0
		MCWB	1.6	4.5	11.7	13.2	18.4	21.5	23.9	22.6	21.2	16.3	10.9	4.1
	5%	DB	2.2	4.7	13.4	20.2	26.1	28.9	31.2	29.1	27.5	20.7	12.7	4.0
		MCWB	0.6	2.4	9.0	11.8	17.0	20.5	22.8	21.3	19.8	14.2	8.5	2.1
	10%	DB	0.8	2.5	10.0	17.6	23.0	27.3	29.4	27.7	25.3	17.6	10.2	2.5
		MCWB	-0.5	0.7	6.5	10.3	15.2	19.3	21.7	20.4	18.8	12.9	6.6	1.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	8.0	15.8	17.4	22.7	24.8	26.8	25.6	23.9	20.4	14.6	8.6
		MCDB	6.7	11.9	19.6	25.0	29.9	31.0	32.0	31.1	29.7	24.4	18.0	10.2
	2%	WB	1.9	4.8	12.3	14.8	19.8	23.1	25.3	24.0	22.4	17.2	11.6	4.3
		MCDB	3.5	7.2	16.0	20.6	26.1	28.8	31.0	28.8	27.5	21.0	15.4	6.2
	5%	WB	0.6	2.6	9.5	12.8	18.3	21.7	24.1	22.9	21.0	15.3	8.8	2.5
		MCDB	2.0	4.6	13.4	18.6	23.8	27.2	29.2	27.7	25.5	19.5	12.3	4.2
	10%	WB	-0.7	0.8	6.5	11.0	16.6	20.4	22.9	21.7	19.7	13.2	6.9	1.1
		MCDB	0.5	2.5	9.6	16.8	21.9	25.4	28.1	26.2	24.5	17.2	9.9	2.5
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.9	9.4	10.8	11.2	10.5	10.7	10.5	10.9	10.0	8.5	7.4
		MCDBR	9.1	10.0	13.4	15.0	15.1	12.5	11.9	11.9	12.9	14.4	12.6	9.5
		MCWBR	7.5	7.5	8.5	8.2	7.1	5.6	5.3	5.2	6.2	8.1	8.3	7.6
	5% WB	MCDBR	8.6	9.5	12.6	13.2	12.9	10.8	10.6	10.3	10.8	11.6	11.7	8.7
		MCWBR	7.3	7.3	8.5	8.1	7.0	5.8	5.6	5.7	6.1	7.7	8.3	7.2
Clear-Sky Solar Irradiance	taub	0.270	0.283	0.320	0.378	0.409	0.400	0.399	0.406	0.377	0.322	0.292	0.269	
	taud	2.412	2.363	2.381	2.289	2.254	2.328	2.354	2.334	2.378	2.527	2.512	2.435	
	Ebn at Noon	856	908	914	881	861	869	865	845	841	854	825	819	
	Edh at Noon	71	91	103	124	133	124	120	118	104	78	65	63	
All-Sky Solar Radiation	RadAvg	1.72	2.81	3.81	4.60	5.33	5.93	6.29	5.24	4.08	2.58	1.65	1.21	
	RadStd	0.20	0.28	0.35	0.46	0.44	0.46	0.23	0.38	0.29	0.34	0.15	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page